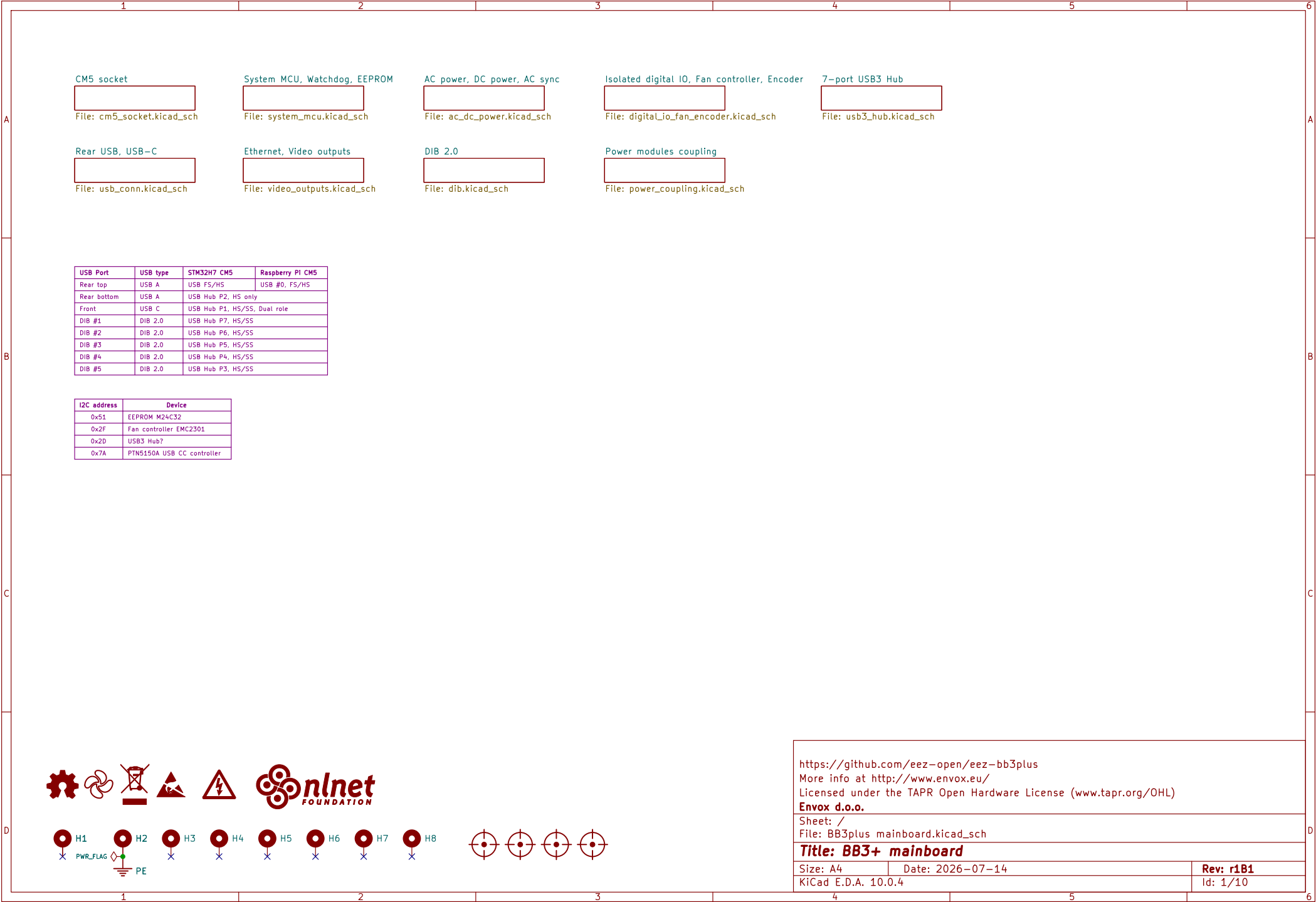
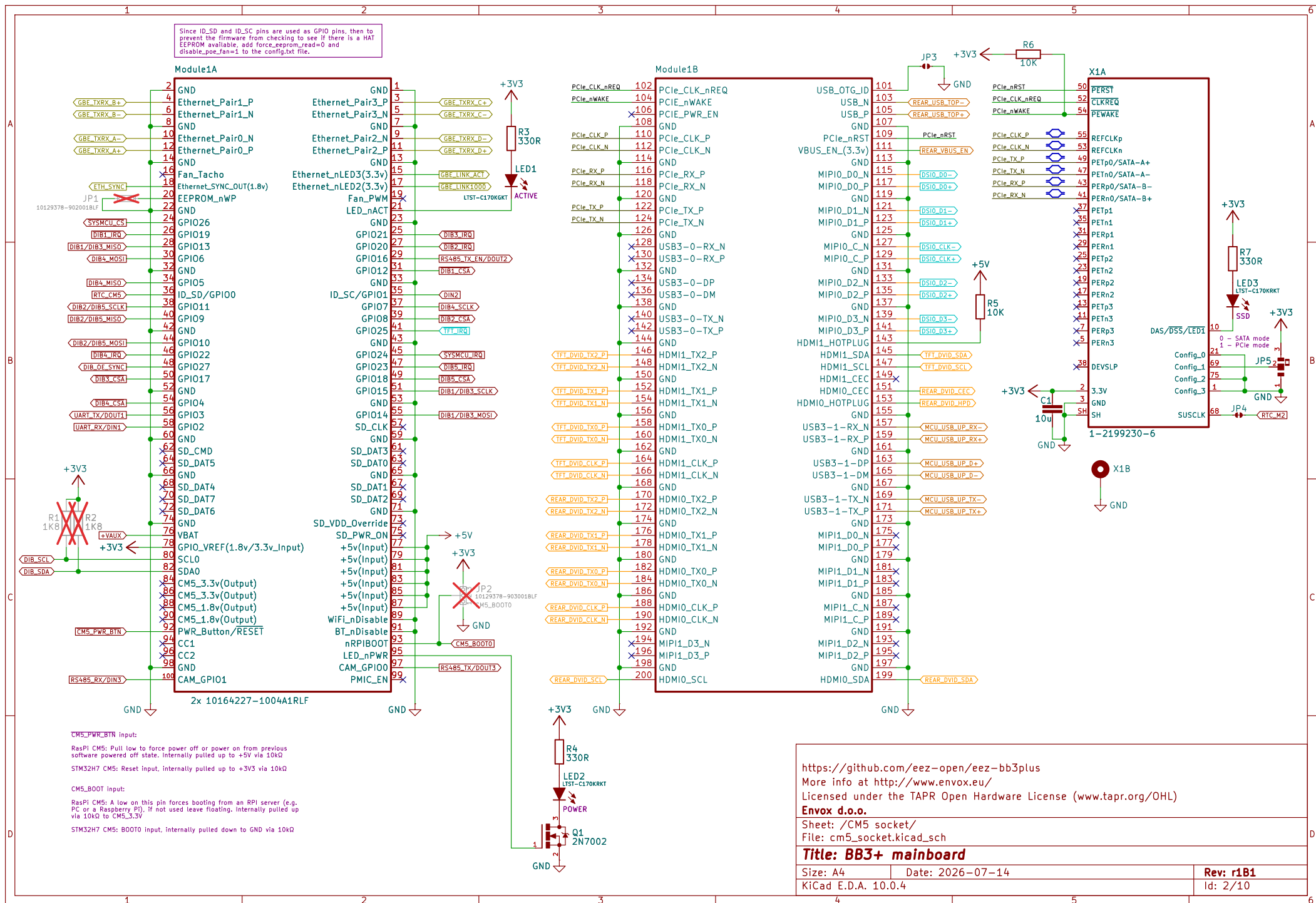






<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /CM5 socket/

File: cm5_socket.kicad_sch

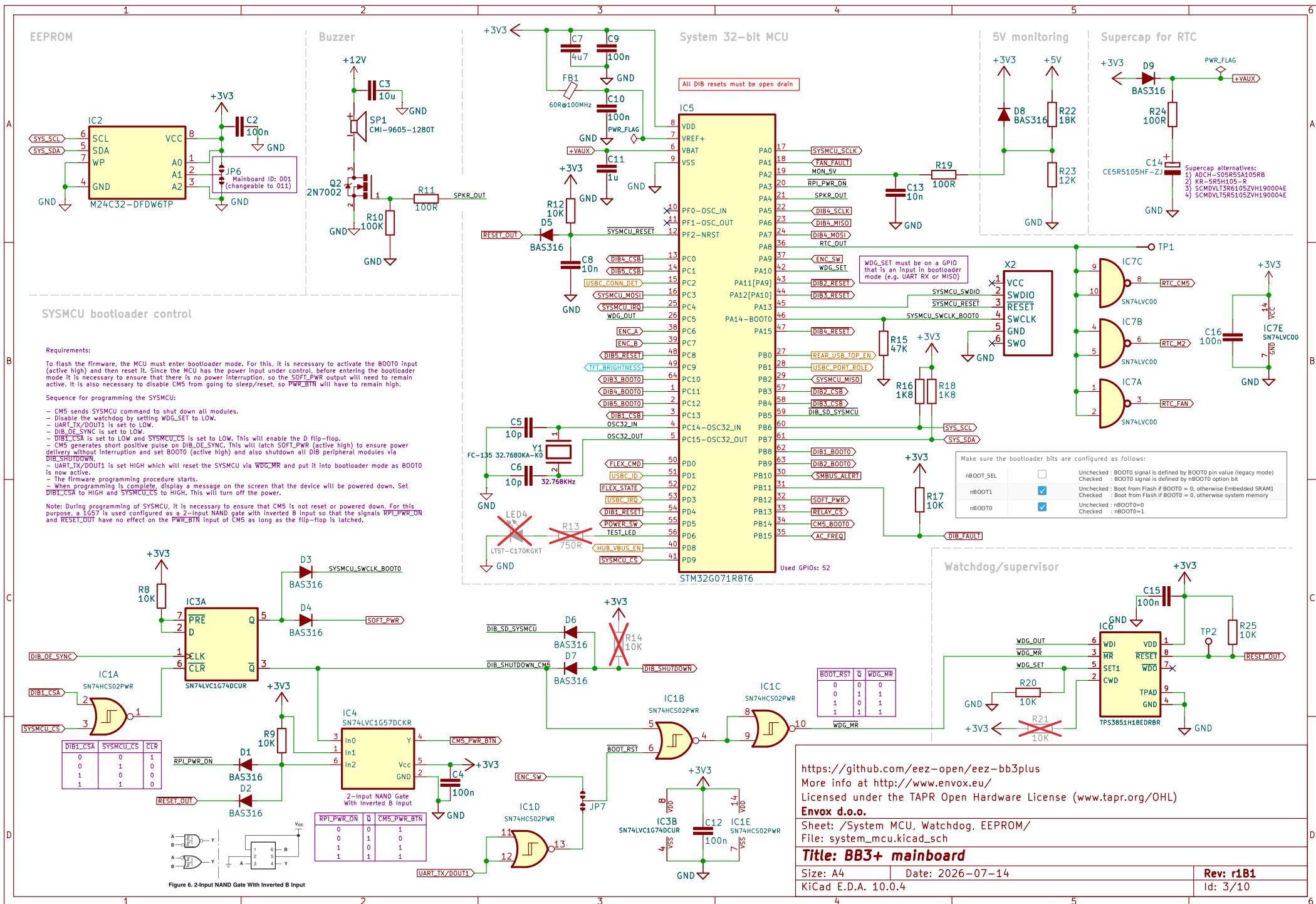
Title: BB3+ mainboard

Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 2/10



<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /System MCU, Watchdog, EEPROM/

File: system_mcu.kicad_sch

Title: BB3+ mainboard

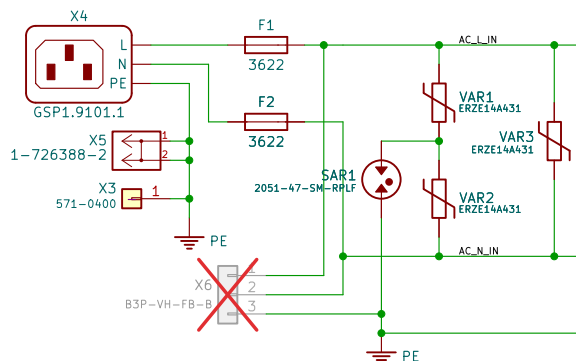
Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

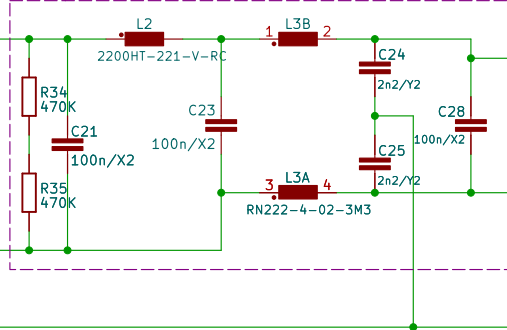
Rev: r1B1

Id: 3/10

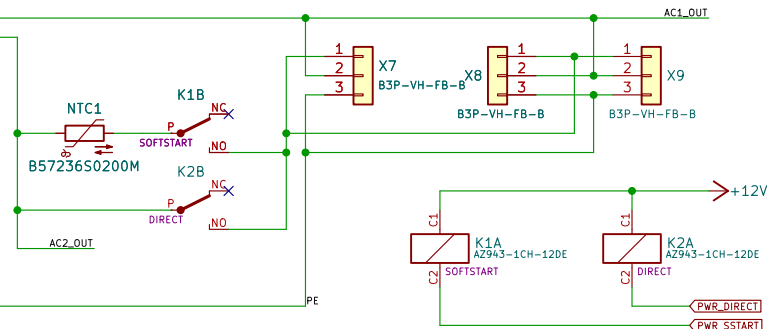
AC inlet, filtering and protection



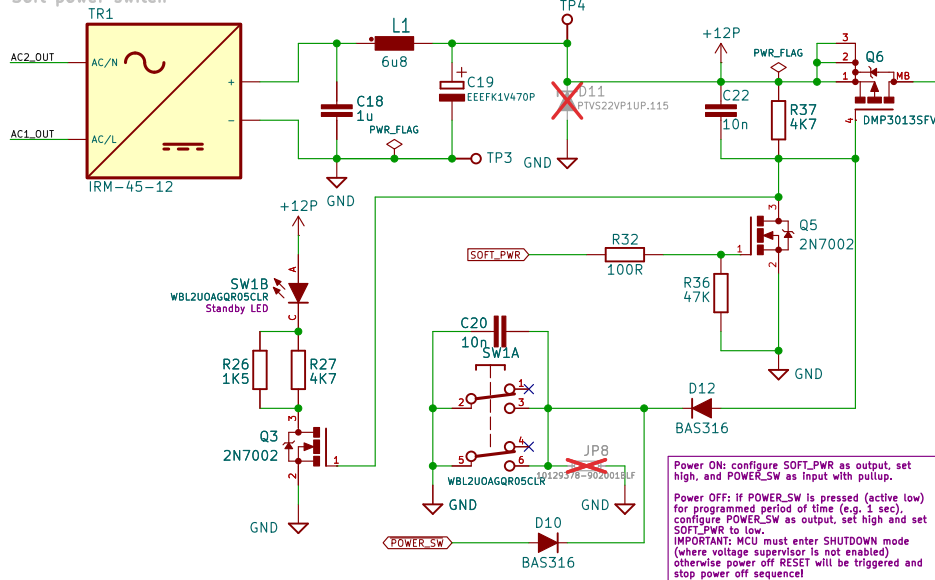
EMI compliance filter



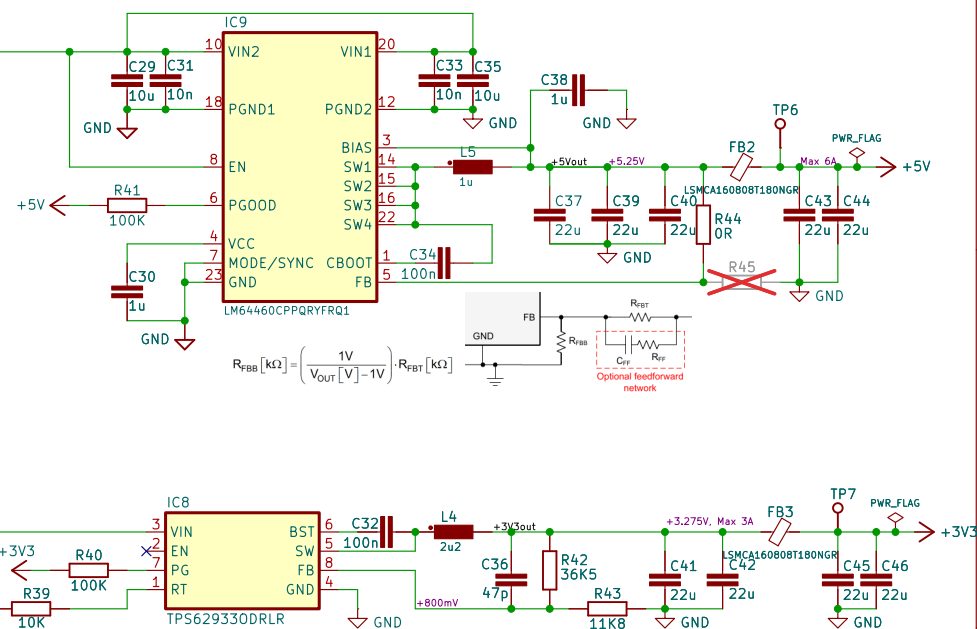
AC power control, Inrush current limiter



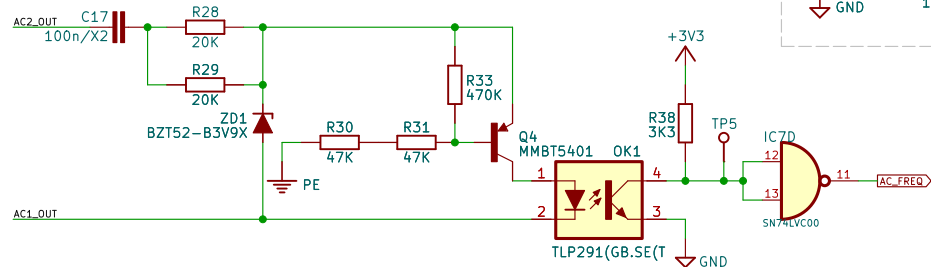
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

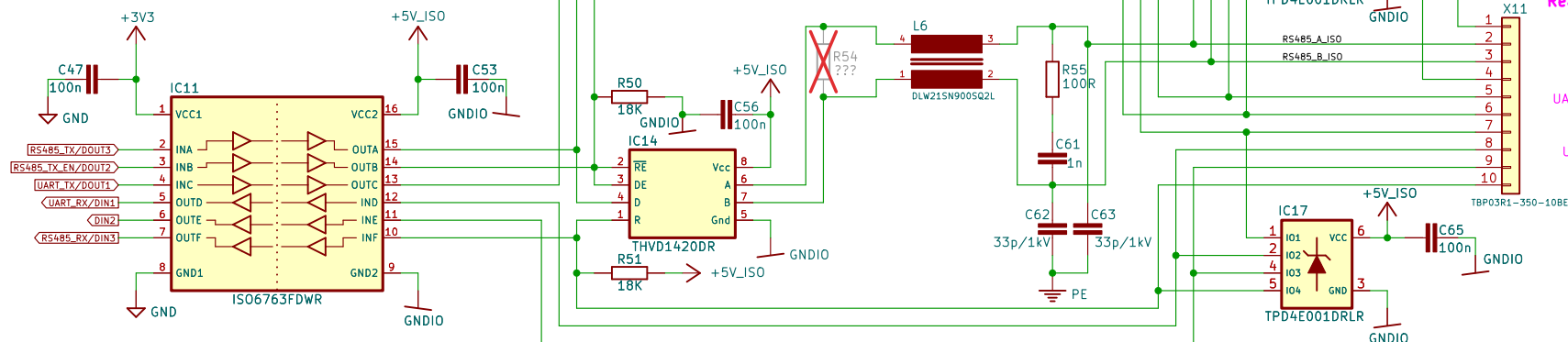
Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 4/10

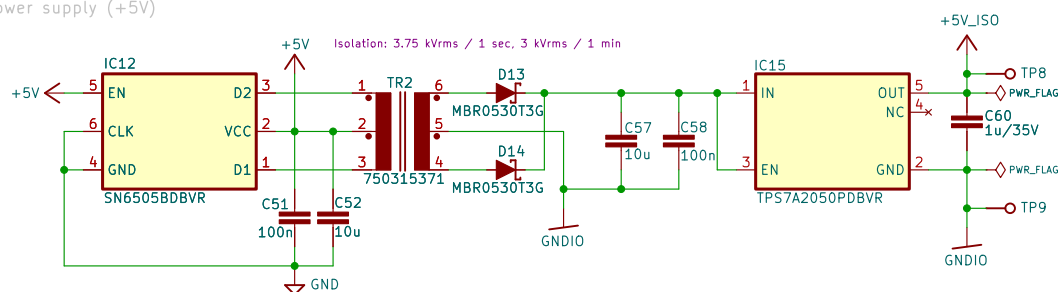
Isolated digital I/O & RS485 transceiver (half-duplex)



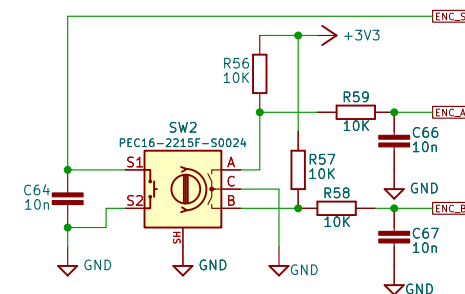
Rear panel I/O connector

- | | |
|---------------|----|
| +5V_ISO | 1 |
| RS485_A_ISO | 2 |
| RS485_A_ISO | 3 |
| GND_ISO | 4 |
| UART_TX/DOUT1 | 5 |
| DOUT2 | 6 |
| DOUT3 | 7 |
| UART_RX/DIN1 | 8 |
| DIN2 | 9 |
| DIN3 | 10 |

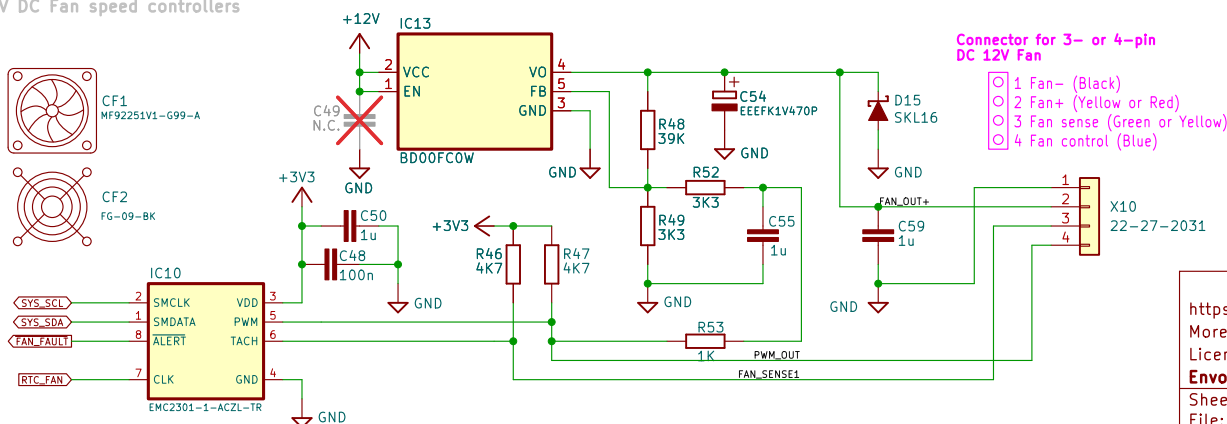
Isolated regulated power supply (+5V)



Encoder with switch



12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

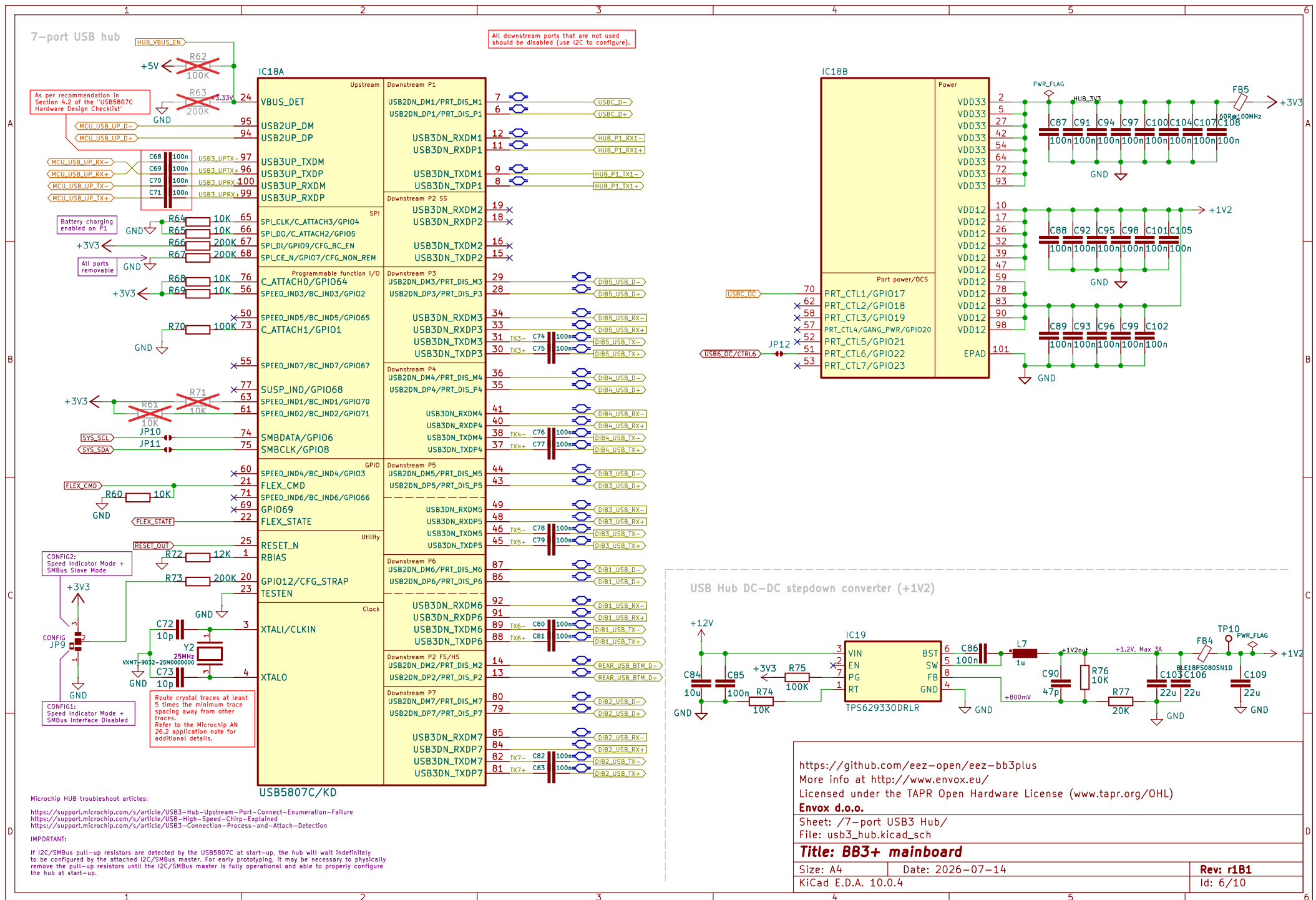
Title: BB3+ mainboard

Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 5/10



IC22: TPS2069E

- IN: 5
- EN: 4
- FLT: 3
- OUT: 1
- GND: 2

IC20: PTN5150AHXMP

- SCL/OUT2: 8
- SDA/OUT1: 7
- ADR/CON_DET: 5
- INTB/OUT3: 6
- ID: 9
- SEL_SEL: 11
- PORT: 3
- VDD: 12
- GND: 10

IC13: USB4056-03-A

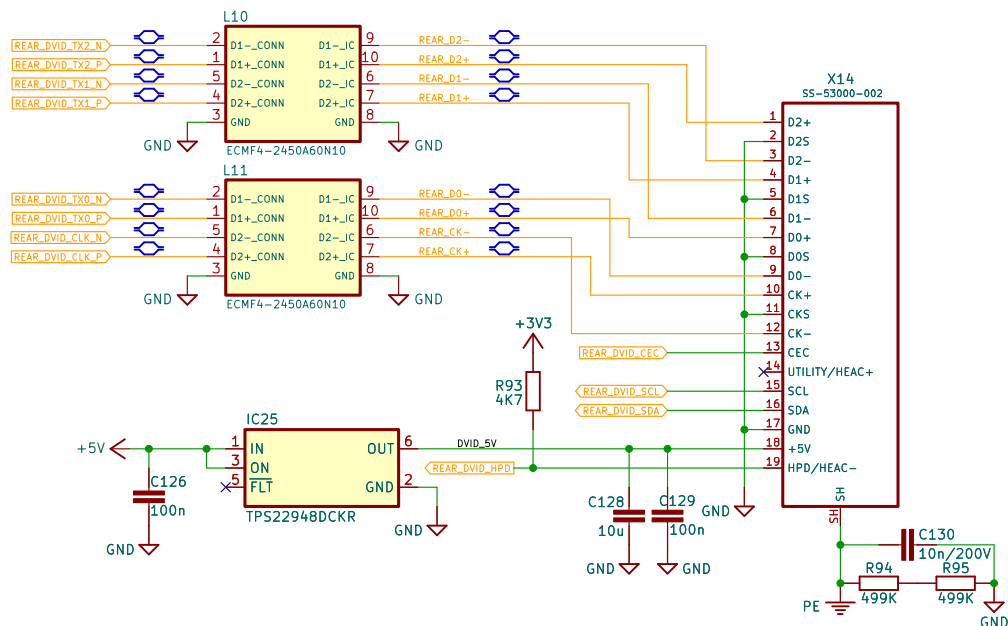
- VBUS_B: B4
- VBUS_A: B9
- SSRXN1: B8
- SSRXN2: B7
- SSRXN3: B6
- SSRXN4: B5
- SSRXN5: B4
- SSRXN6: B3
- SSRXN7: B2
- SSRXN8: B1
- SSRXN9: A9
- SSRXN10: A8
- SSRXN11: A7
- SSRXN12: A6
- SSRXN13: A5
- SSRXN14: A4
- SSRXN15: A3
- SSRXN16: A2
- SSRXN17: A1
- SSRXN18: A0
- SSRXN19: A12
- SSRXN20: A11
- SSRXN21: A10
- SSRXN22: A9
- SSRXN23: A8
- SSRXN24: A7
- SSRXN25: A6
- SSRXN26: A5
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- SSRXN28: A3
- SSRXN29: A2
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- SSRXN36: A8
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- SSRXN44: A0
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- SSRXN241: A11
- SSRXN242: A

[illegible]

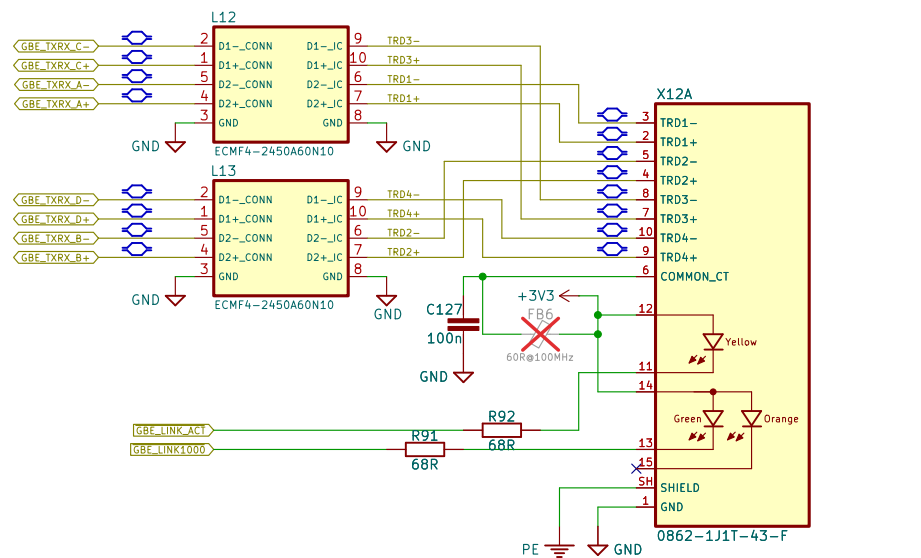
KiCad E.D.A. 10.0.4

18. 7/10

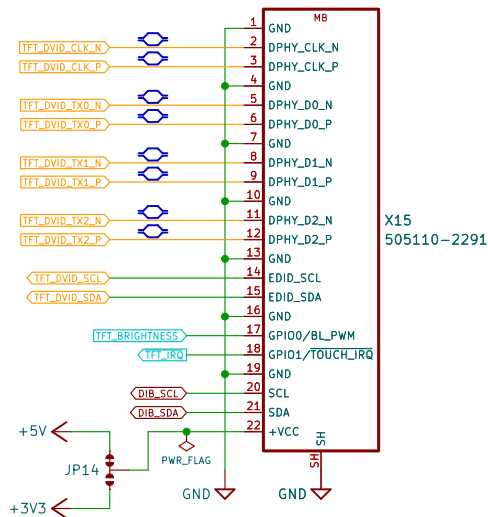
External monitor digital output



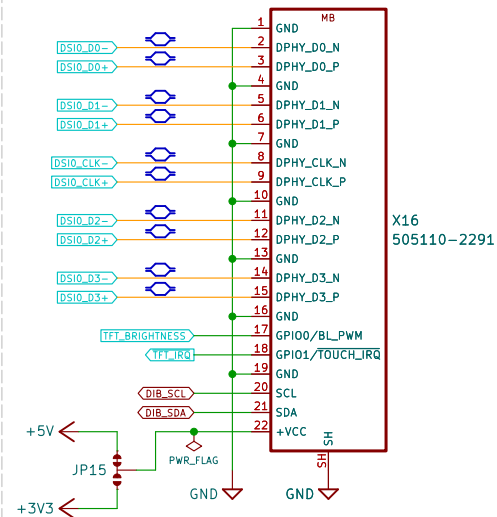
Ethernet RJ45+mag



TFT Touchscreen display DVID output (22-pin FFC)



TFT Touchscreen display DSI output (22-pin FFC)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /Ethernet, Video outputs/

File: video_outputs.kicad_sch

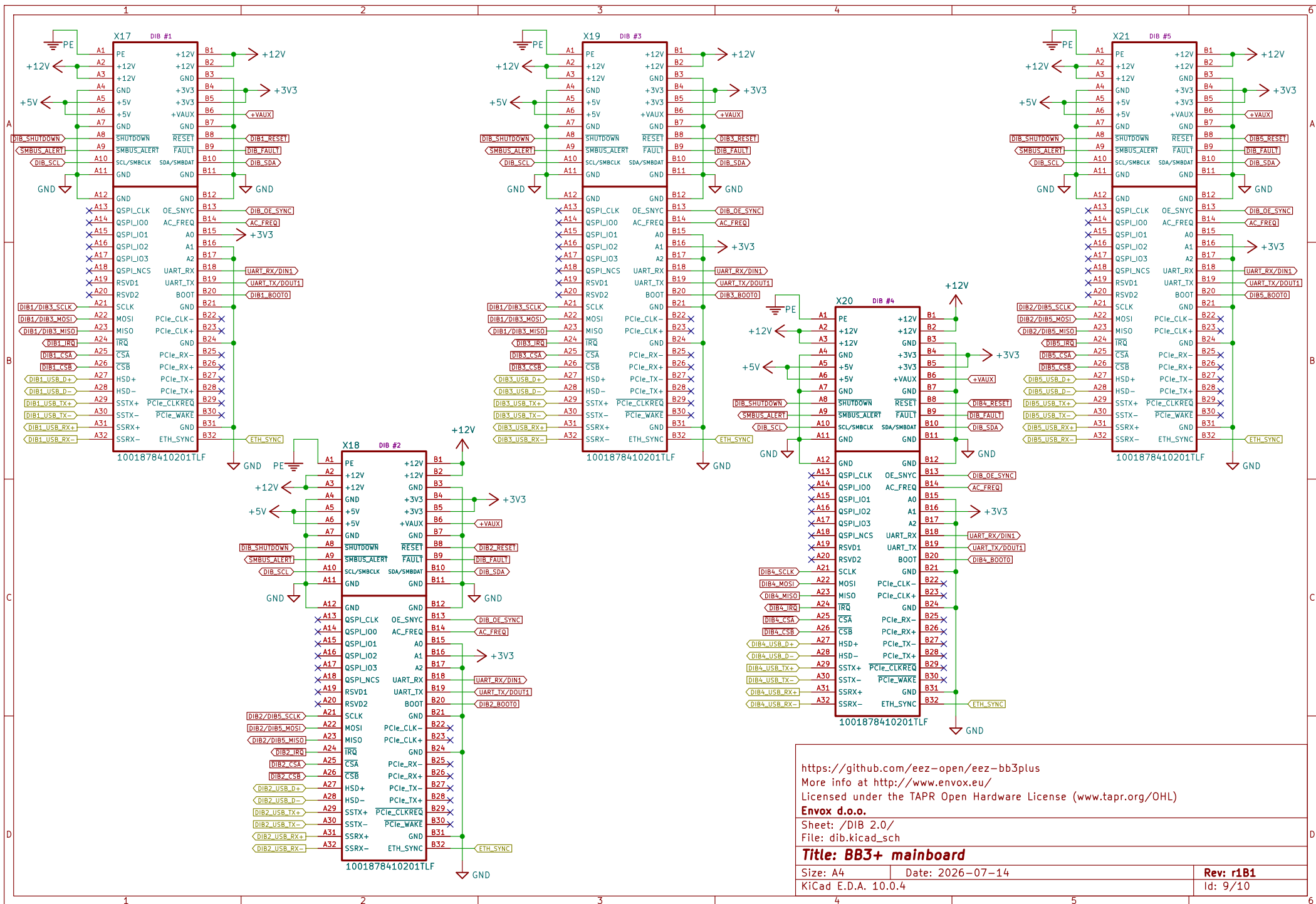
Title: BB3+ mainboard

Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 8/10

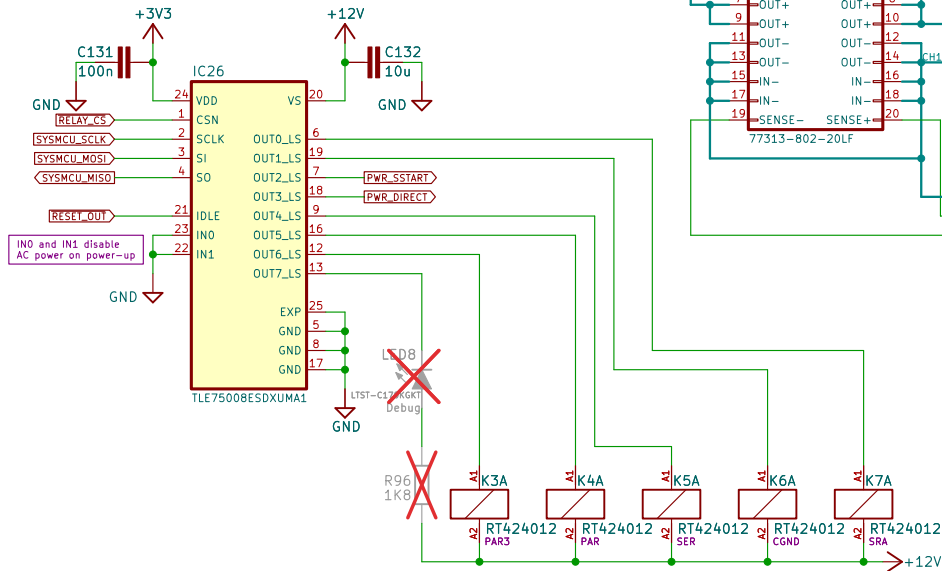


<https://github.com/eez-open/eez-bb3plus>
 More info at <http://www.envox.eu/>
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Envox d.o.o.
 Sheet: /DIB 2.0/
 File: dib.kicad_sch
Title: BB3+ mainboard
 Size: A4 Date: 2026-07-14
 KiCad E.D.A. 10.0.4 Rev: r1B1
 Id: 9/10

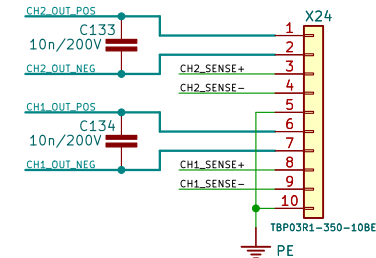
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ IN
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 2	3
CH2_SENSE+ 3	4
CH2_SENSE- 4	5
PE 5	6
CH1_OUT+ 6	7
CH1_OUT- 7	8
CH1_SENSE+ 8	9
CH1_SENSE- 9	10
PE 10	11

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-07-14

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 10/10